**University of Engineering & Management, Kolkata**
Institute of Engineering and Management, Newtown Campus
Department of Computer Applications

# ASSIGNMENT SUBMISSION — TOP SHEET

Technical Class Assessment — April 2026
Database Management Systems & Java with JDBC | MCA Programme

## STUDENT DETAILS

| | | | |
|---|---|---|---|
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| **Section** | A | **Semester** | 2 |
| **Programme** | MCA | **Academic Year** | 2025-2026 |
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## ASSIGNMENT DETAILS

| | | | |
|---|---|---|---|
| **Subject** | Java – DBMS Assignment | **Set Number** | 3 |
| **Problem Title** | The Canteen Queue | | |
| **Language Used** | DBMS & Core Java | **Database Used** | MySQL 8.0.44 |
| **Submission Date** | 10TH APRIL 2026 | **Total Pages** | 12 |

## SUBMISSION CHECKLIST

| # | Item | Included ( / ) | Pages / Remarks |
|---|---|---|---|
| 1 | Printed Java Source Code (DBConnection.java) | | |
| 2 | Printed Java Source Code (Main.java) | | |
| 3 | Printed SQL Schema File (schema.sql) | | |

| | |
|---|---|
| 5 | Problem Statement (this top sheet attached) |

## DECLARATION

*I hereby declare that this assignment is my own original work. I have not copied from any other student or used any unauthorised sources. I understand that plagiarism will result in zero marks and disciplinary action.*

ANTORI DAS (10.04.2026)

*Student Signature & Date*

*Faculty Signature & Date*

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**TECHNICAL CLASS ASSESSMENT — April – 2026**

| Degree: MCA | Stream: MCA |
|---|---|
| Due Date: _____ | Set No: 3 |

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# The Canteen Queue

*[Domain: Campus Life j*

| | |
|---|---|
| **Language** | Core Java (Console Application) |
| Database | MySQL |
| **Connectivity** | JDBC (Java Database Connectivity) |

## The Story

> *"If only we knew* /?ovr *many students want rice and how many want noodles — before 12 o'clock."*

*The college canteen at IEM Newtown sees a lunch-hour rush every day. Students crowd the counter, orders get mixed up, and the cook has no idea how many plates of each dish to prepare. Food is either wasted or runs out too early.*

*Your task is to build a Canteen Order Management System using Java and MySQL.*

## Your Task

*Build a Canteen Order Management System as a Java console application using JDBC and MySQL.*

### Your system must allow:

- Add a new menu item with name and price
- Place an order for a student (student name, item, quantity)
- Mark an order as served once food is handed over
- Display total revenue and most ordered item of the day

## Database Schema

*Create the following table(s) in MySQL:*

```
CREATE TABLE menu_items (
    item_id   INT PRIMARY KEY AUTO_INCREMENT,
    item_name VARCHAR(100),
    price     DECIMAL(6,2)
```

```
CREATE TABLE orders (
    order_id    INT PRIMARY KEY AUTO_INCREMENT,
    student     VARCHAR(100),
    item id     INT,
    quantity    INT DEFAULT 1,
    ordered at DATETIME DEFAULT NOW(),
    served      BOOLEAN DEFAULT FALSE,
    FOREIGN KEY (item id) REFERENCES menu items(item_id)
```

## Required Features

| Feature | Implementation Hint |
|---|---|
| Add Menu Item | INSERT into menu_items with name and price |
| Place Order | INSERT into orders with student, item id, quantity |
| Mark Served | UPDATE orders SET served=TRUE WHERE order id=? |
| Daily Report | SUM(price”quantity) for revenue; ORDER BY COUNT(*) for top item |

## Constraints

- Use Core Java only — no Spring, Hibernate, or web frameworks
- Connect to MySQL using JDBC (no JPA / ORM)
- All operations must be menu-driven via console
- No GUI required — Scanner-based input is sufficient
- Handle SQL exceptions with try-catch blocks
- Use PreparedStatement for all SQL queries

## Sample Interaction

```
        C    e   ystem =====
 1  Add Mene :t:
        e
 :  M  l ORd   Served

      i
 EntE  choice: 4
 Today's Revenue: Rs 2340.00
 Most Ordered: Chicken Rice (47 plates)
```

## Submission Checklist

- DBConnection.java (JDBC connection class)
- Main.java (menu-driven main program)
- SQL file: schema.sql (CREATE TABLE statements)
- screenshot.png (terminal output showing at least 2 features working)

*“A database is not just data. It is the memory of your application. Make it reliable.”*

## SRC/MAIN.JAVA

```
package src;

import java.sql.*;
import java.util.Scanner;

public class Main {
   static Scanner sc = new Scanner(System.in);

   public static void main(String[] args) {

      while (true) {
         System.out.println("\n===== Canteen System =====");
         System.out.println("1. Add Menu Item");
         System.out.println("2. Place Order");
         System.out.println("3. Mark Order Served");
         System.out.println("4. Daily Report");
         System.out.println("5. Exit");

         System.out.print("Enter choice: ");
         int choice = sc.nextInt();

         switch (choice) {
            case 1:
               addMenuItem();
               break;
            case 2:
               placeOrder();
               break;
            case 3:
               markServed();
               break;
            case 4:
               dailyReport();
               break;
            case 5:
               System.exit(0);
         }
      }
   }

   public static void addMenuItem() {
      try (Connection con = DBConnection.getConnection()) {

         sc.nextLine();
         System.out.print("Enter item name: ");
         String name = sc.nextLine();

         System.out.print("Enter price: ");
         double price = sc.nextDouble();

         String sql = "INSERT INTO menu_items (item_name, price) VALUES (?, ?)";
         PreparedStatement ps = con.prepareStatement(sql);
         ps.setString(1, name);
         ps.setDouble(2, price);
```

```
        ps.executeUpdate();
        System.out.println("Item Added!");

    } catch (Exception e) {
        e.printStackTrace();
    }
}

public static void placeOrder() {
    try (Connection con = DBConnection.getConnection()) {

        sc.nextLine();
        System.out.print("Enter student name: ");
        String student = sc.nextLine();

        Statement st = con.createStatement();
        ResultSet rs = st.executeQuery("SELECT * FROM menu_items");

        System.out.println("Menu:");
        while (rs.next()) {
            System.out.println(rs.getInt(1) + " " + rs.getString(2) + " Rs" + rs.getDouble(3));
        }

        System.out.print("Enter item ID: ");
        int id = sc.nextInt();

        System.out.print("Enter quantity: ");
        int qty = sc.nextInt();

        String sql = "INSERT INTO orders (student, item_id, quantity) VALUES (?, ?, ?)";
        PreparedStatement ps = con.prepareStatement(sql);
        ps.setString(1, student);
        ps.setInt(2, id);
        ps.setInt(3, qty);

        ps.executeUpdate();
        System.out.println("Order Placed!");

    } catch (Exception e) {
        e.printStackTrace();
    }
}

public static void markServed() {
    try (Connection con = DBConnection.getConnection()) {

        System.out.print("Enter Order ID: ");
        int id = sc.nextInt();

        String sql = "UPDATE orders SET served = TRUE WHERE order_id=?";
        PreparedStatement ps = con.prepareStatement(sql);
        ps.setInt(1, id);

        int rows = ps.executeUpdate();

        if (rows > 0)
```

```
            System.out.println("Order Served!");
        else
            System.out.println("Not Found!");

    } catch (Exception e) {
        e.printStackTrace();
    }
}

public static void dailyReport() {
    try (Connection con = DBConnection.getConnection()) {

        String rev = "SELECT SUM(m.price * o.quantity) FROM orders o JOIN menu_items m ON
o.item_id = m.item_id";
        ResultSet rs1 = con.createStatement().executeQuery(rev);

        if (rs1.next())
            System.out.println("Revenue: Rs " + rs1.getDouble(1));

        String top = "SELECT m.item_name, SUM(o.quantity) total FROM orders o JOIN menu_items m
ON o.item_id=m.item_id GROUP BY m.item_name ORDER BY total DESC LIMIT 1";
        ResultSet rs2 = con.createStatement().executeQuery(top);

        if (rs2.next())
            System.out.println("Top Item: " + rs2.getString(1) + " (" + rs2.getInt(2) + ")");

    } catch (Exception e) {
        e.printStackTrace();
    }
}
}
```

## SRC/DBCONNECTION.JAVA

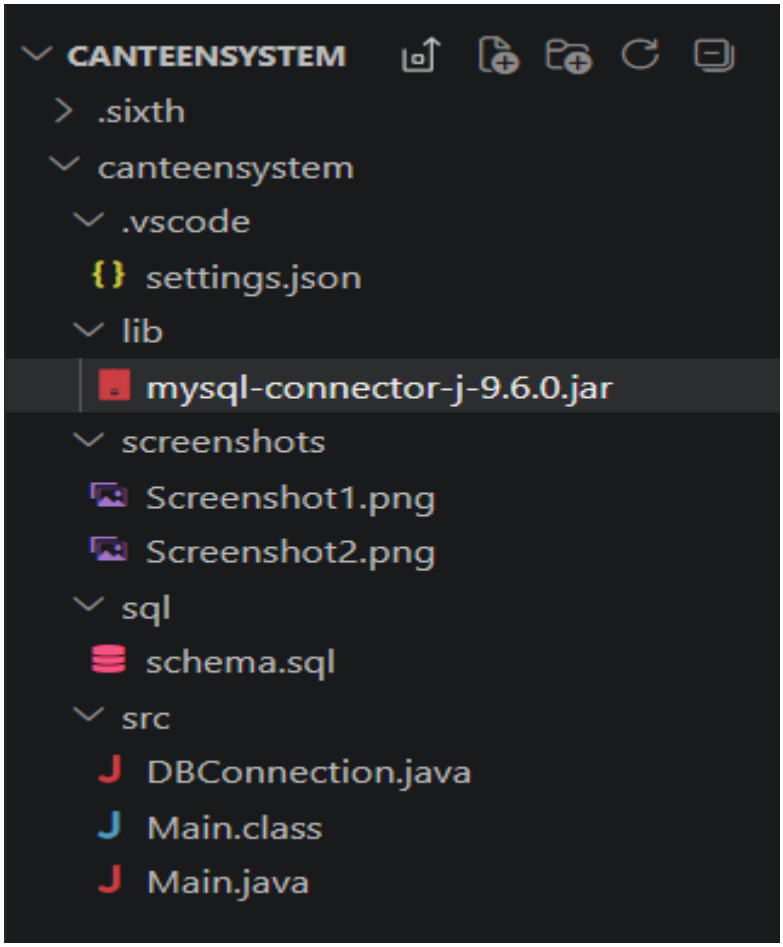
```
package src;

import java.sql.*;

public class DBConnection {

   static final String URL =
"jdbc:mysql://localhost:3306/canteen_db?useSSL=false&allowPublicKeyRetrieval=true&serverTimez
one=UTC";
   static final String USER = "testuser";
   static final String PASSWORD = "1234";

   public static Connection getConnection() {
      try {
         return DriverManager.getConnection(URL, USER, PASSWORD);
      } catch (SQLException e) {
         System.out.println("Database Connection Failed!");
         e.printStackTrace();
         return null;
      }
   }
}
```

## SQL/SCHEMA.SQL

```sql
CREATE DATABASE canteen_db;

USE canteen_db;

CREATE TABLE menu_items (
   item_id INT PRIMARY KEY AUTO_INCREMENT,
   item_name VARCHAR(100),
   price DECIMAL(6,2)
);

CREATE TABLE orders (
   order_id INT PRIMARY KEY AUTO_INCREMENT,
   student VARCHAR(100),
   item_id INT,
   quantity INT DEFAULT 1,
   ordered_at DATETIME DEFAULT NOW(),
   served BOOLEAN DEFAULT FALSE,
   FOREIGN KEY (item_id) REFERENCES menu_items(item_id)
);
```

# SCREENSHOTS

```
PS C:\canteensystem> javac -cp ".;lib/mysql-connector-j-9.6.0.jar" *.java
PS C:\canteensystem> java -cp ".;lib/mysql-connector-j-9.6.0.jar" Main

===== Canteen System =====
1. Add Menu Item
2. Place Order
3. Mark Order Served
4. Daily Report
5. Exit
Enter choice: 1
Enter item name: samosa
Enter price: 10.0
Item Added!

===== Canteen System =====
1. Add Menu Item
2. Place Order
3. Mark Order Served
4. Daily Report
5. Exit
Enter choice: 2
Enter student name: Ankita
Menu:
1 samosa Rs10.0
Enter item ID: 1
Enter quantity: 3
Order Placed!

===== Canteen System =====
1. Add Menu Item
2. Place Order
3. Mark Order Served
4. Daily Report
5. Exit
Enter choice: 1
Enter item name: Biryani
Enter price: 80.0
Item Added!
```

```
===== Canteen System =====
1. Add Menu Item
2. Place Order
3. Mark Order Served
4. Daily Report
5. Exit
Enter choice: 2
Enter student name: Ankit
Menu:
1 samosa Rs10.0
2 Biryani Rs80.0
Enter item ID: 2
Enter quantity: 3
Order Placed!

===== Canteen System =====
1. Add Menu Item
2. Place Order
3. Mark Order Served
4. Daily Report
5. Exit
Enter choice: 3
Enter Order ID: 2
Order Served!

===== Canteen System =====
1. Add Menu Item
2. Place Order
3. Mark Order Served
4. Daily Report
5. Exit
Enter choice: 4
Revenue: Rs 270.0
Top Item: samosa (3)
```

## ***FILE STRUCTURE***